Development: Chinese–English Idiom-Aware Translation System

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Existing machine translation systems perform well on general text but often fail to correctly interpret idiomatic expressions due to their non-literal and cultural meanings. In this project, we develop a Chinese–English idiom-aware translation system that combines translation APIs with embedding-based idiom retrieval. The system detects idioms in input text, generates context-aware translations, and provides short cultural explanations to improve user understanding. We implement the system as a web application, and evaluate its performance against standard translation APIs.

The link to the project repository can be found here:
https://github.com/zchnee/CS510-final-project

CCS Concepts: • **Information systems → Information retrieval**; • **Computing methodologies → Machine translation**.

Additional Key Words and Phrases: Chinese-English translation, idiom translation, contextual translation, FAISS, semantic retrieval, machine translation, cultural understanding, web application

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1 Introduction

Machine translation systems have become increasingly important for multilingual communication, language learning, and cross-cultural interaction. Modern translation tools such as Google Translate achieve strong performance on general sentence translation through neural language models and large-scale training datasets. However, these systems often struggle with idiomatic expressions because idioms are highly contextual and culturally dependent. Literal translation of idioms may produce grammatically correct but semantically incorrect results, leading to misunderstanding and loss of cultural meaning.

This issue is particularly significant in Chinese–English translation. Many Chinese idioms contain condensed historical stories or cultural references that cannot be directly translated word-for-word. Existing translation systems usually provide literal translations without sufficient contextual explanation, making it difficult for language learners and non-native speakers to fully understand the intended meaning.

To address this problem, we propose a Chinese–English idiom-aware translation system that combines translation APIs with embedding-based idiom retrieval and contextual explanation generation. Our system detects idiomatic expressions within input text, retrieves semantically related idioms using FAISS similarity search, and provides short cultural explanations alongside the translated result. The system is implemented as a web application using a React frontend and a Node.js/FastAPI backend.

The main contributions of this project include:

- A context-aware Chinese–English idiom translation pipeline
- An embedding-based idiom retrieval system using FAISS
- Contextual idiom explanations to improve cultural understanding
- A web-based interactive translation application

2 Software Overview

Our project develops a Chinese–English idiom-aware translation system designed to improve contextual understanding of idiomatic expressions. The system combines the *@vitalets/google-translate-api* translation library with embedding-based idiom retrieval and contextual explanation retrieval from a curated dataset. Unlike traditional translation systems that produce literal output, our approach focuses on preserving the semantic and cultural meanings of idiomatic expressions.

2.1 Motivation

Existing machine translation systems perform well on general sentence translation but often fail to correctly interpret idiomatic expressions. Chinese idioms are highly contextual and frequently contain condensed cultural or historical meanings that cannot be translated literally. As a result, users may receive translations that are grammatically correct but semantically misleading.

Our project addresses this problem by augmenting standard machine translation with idiom detection and cultural annotation, providing users with the intended meaning, alternative translations, and contextual examples for each identified idiomatic expression.

2.2 Intended Users

The system is designed for users who frequently interact with bilingual Chinese–English content. Target users include:

- Language learners studying Chinese or English as a second language
- Bilingual speakers seeking deeper cultural understanding
- Translators and educators requiring contextual idiom interpretation

The system is particularly useful in educational and intercultural communication scenarios where understanding idiomatic meaning is essential.

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2.3 Software Architecture

The overall system consists of four major components:

- Frontend user interface implemented with React, TypeScript, and Vite
- Backend API server built with FastAPI (Python)
- Translation module using a Node.js bridge to `@vitalets/google-translate-api`
- Offline FAISS vector index built from multilingual sentence embeddings for idiom retrieval

The system operates in two phases. In the offline phase, `build_index.py` loads the aligned idiom dataset, generates sentence embeddings using the `paraphrase-multilingual-MiniLM-L12-v2` sentence-transformer model, and stores the resulting vectors in a FAISS index on disk alongside a metadata pickle file. This step runs once and does not affect runtime performance.

In the online phase, the FastAPI server loads the idiom dataset into memory at startup, making all idiom entries available for constant-time lookup without repeated disk reads. When a user submits a sentence, the backend simultaneously issues a translation request through the Node.js bridge and scans the input for idiomatic spans using sliding-window dictionary matching against the preloaded idiom set. Each detected idiom is enriched with its primary meaning, alternative translations, and a bilingual example sentence retrieved from the in-memory dataset. The translated result and idiom annotations are returned to the React frontend and rendered as highlighted spans with expandable explanation cards.

3 Implementation Details

3.1 Major Functions

The backend server provides several API endpoints to support idiom-aware Chinese–English translation.

- `GET /`
  This endpoint is used to check whether the backend server is up and running. It returns a simple server status message.
- `POST /detect`
  This endpoint detects Chinese idioms in the input text. It takes a Chinese sentence as input and returns the detected idioms with their positions in the sentence, English meanings, alternative English meanings, and example Chinese–English sentence pairs.
- `POST /translate`
  This endpoint translates Chinese text into English. It sends the input text to a Node.js translation bridge, which uses `@vitalets/google-translate-api`, and returns the translated result.

Together, these endpoints support the main workflow of the system. The frontend first sends the user input to the backend for translation and idiom detection. The detected idiom metadata is then displayed with the translated sentence, allowing users to understand both the literal translation and the contextual meaning of idiomatic expressions.

3.2 Idiom Dataset Buildup

The idiom dataset used in our system was built by aligning two publicly available resources: PETCI [Tang 2022] and IdiomTranslate30[Tang et al. 2024]. PETCI provides Chinese idioms with English meanings collected from dictionaries and mainstream translation APIs. IdiomTranslate30 provides idioms with Chinese example sentences and English sentence-level translations. By combining these two sources, our dataset includes both idiom-level meanings and sentence-level usage examples.

The final aligned dataset is stored in JSON format. Each entry contains the Chinese idiom, its primary English meaning from PETCI, alternative English meanings, a Chinese example sentence from IdiomTranslate30, the corresponding English sentence translation, and metadata used for retrieval.

For each idiom entry, the dataset stores:

- `idiom_zh`: the original Chinese idiom
- `meaning_en`: the primary English meaning from PETCI
- `alternative_meanings_en`: alternative English meanings
- `sentence_zh`: Chinese example sentence from IdiomTranslate30
- `sentence_en`: English translation of the example sentence
- `embedding_text_zh`: Chinese text used for sentence embedding and FAISS retrieval
- `petci_id`: the corresponding PETCI entry ID
- `idiomtranslate30_index`: the corresponding IdiomTranslate30 index

Before retrieval, the dataset is processed by `get_data.py` and then indexed by `build_index.py`. Sentence embeddings are generated for the Chinese embedding text and stored in a FAISS index, allowing the backend to efficiently retrieve idiom metadata during detection and translation.

4 Usage Demonstration

4.1 Basic Translation with Idiom Detection

A user begins by entering a Chinese sentence into the input field and submitting the query. The system simultaneously issues a `POST /translate` request to obtain the base English translation and a `POST /detect` request to identify idiomatic spans in the original text. Both requests resolve before the output is rendered.

Consider the input sentence containing two idioms: 他们威逼利诱，我却半信半疑，不知道该相信谁。 The system produces the base English translation alongside highlighted idiom spans corresponding to the two detected expressions in the original Chinese text (See Figure 1). Each highlighted span is interactive — clicking or hovering over a span expands an explanation card beneath the translation panel.

4.2 Idiom Explanation Cards

For each detected idiom, the system renders an explanation card containing the idiom's primary English meaning, a list of alternative translations, and a bilingual example sentence drawn from the aligned dataset. For the sentence above, two cards are rendered.

For 威逼利诱, the card displays the primary meaning "coercion and coaxing," with alternatives including "combine threats with inducements" and "alternate intimidation and bribery."

For 半信半疑, the card displays the primary meaning "not quite convinced," with alternatives including "half believe and half doubt" and "be dubious of."

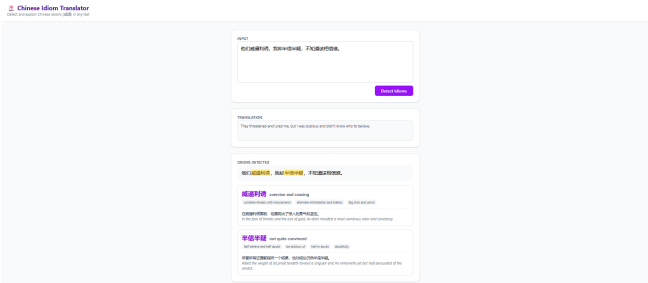

Fig. 1. Translation output with two detected idioms highlighted and explanation cards rendered.

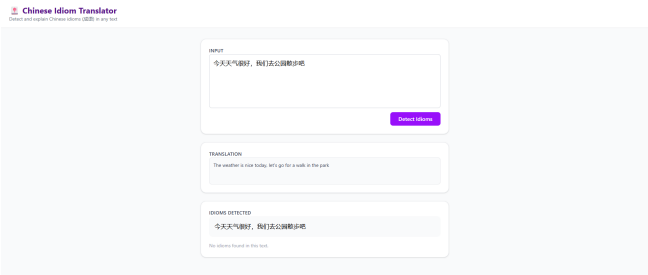

Fig. 2. Translation output for a sentence containing no idiomatic expressions.

4.3 Sentences Without Idioms

When the user submits a sentence containing no idiomatic expressions — for example, 今天天气很好， 我们去公园散步吧 — the system returns only the base translation with no highlighted spans and no explanation cards, correctly indicating that no idioms were detected (See Figure 2).

4.4 API Access

The backend exposes two endpoints, `POST /detect` and `POST /translate`, both accepting JSON request bodies. For users wishing to interact with the backend directly, the system exposes an auto-generated interactive API explorer where both endpoints can be tested with arbitrary Chinese input without requiring the frontend to be running.

The `POST /detect` endpoint accepts a JSON body with a single `text` field containing the Chinese input string. It returns a list of enriched idiom matches, where each entry contains the matched idiom, its character span indices (`start`, `end`), its primary English meaning, a list of alternative translations, and a bilingual example sentence pair. If the input text is empty, the endpoint returns a `400 Bad Request` error.

The `POST /translate` endpoint accepts a JSON body with a `text` field and optional `to` and `source` language fields, defaulting to English output and automatic source detection respectively. The endpoint delegates translation to the Node.js bridge via a subprocess call with a 20-second timeout, returning the translated string on success. The endpoint handles three failure cases explicitly: a `500` error if the Node.js runtime is unavailable, a `504` error if the translation request times out, and a `502` error if the translation provider returns an invalid response.

Both endpoints validate that the input text is non-empty before processing, and the idiom dataset is loaded once into memory at server startup rather than on each request, ensuring consistent low-latency responses for the `/detect` endpoint regardless of dataset size.

5 Evaluation Results

5.1 Setup

The publicly available dataset ChengyuSTS [Qiu et al. 2025] was used as a test set to evaluate our translator, due to its dataset being distinct from our dataset that was sourced from PETCI and Idiom-Translate30. In particular, each line in the JSON file was truncated and had both its "idiom" and "sent" value extracted, which contained both a Chinese idiomatic phrase and the sentence that the associated idiom was written in, respectively. Some sentences did not actually have an idiom within them; as such the "idiom" value would simply be `null`.

Next, we set our benchmark definition for each idiom to be determined by OpenAI's GPT-5.4 model, as its competitive performance in Chinese-to-English translation tasks makes it an excellent LLM-as-data-annotator. Each idiom/sentence pair from the ChengyuSTS dataset would be fed into our `/detect` and `/translate` endpoints and be compared against the benchmark.

The process of running our endpoints to check if an idiom can be detected in its given sentence can be automated, but we decided to manually compare the benchmark's definition of an idiom with the definitions given by our `/translate` endpoint, as automated evaluation could possibly lack the pragmatic sensitivity required to distinguish between literal translations and figurative interpretations - human-in-the-loop input is necessary to make sure that the nuance and cultural intent that idioms and figurative language often display are not misrepresented. This manual evaluation is only applied to sentences that have a detected idiom within them and are passed to the human expert for further review.

5.2 Metrics

To more specifically define the metrics that were hinted above, the first set of metrics that we are observing are the detection metrics, more specifically, we are distinguishing:

- Sentences that have successfully detected idioms
- Sentences that are successfully identified to contain no idioms
- Sentences that contain an idiom that isn't detected or contain phrases that are incorrectly identified as idioms

- Sentences that do not contain an idiom but are claimed to have one by the translator

The second set of metrics that we are observing are the definition metrics, which we are distinguishing from the manual evaluation from the human expert. The following labels are:

- Approved: Most definitions are synonymous with the benchmark's definitions, or at least one definition exactly matches that of the benchmark's.
- Unsatisfactory: Definitions only at best approximate what the benchmark may define instead of being synonymous, or are excessively roundabout in meaning; definitions may refer to the literal meaning of the idiom if it was similar to the figurative meaning.
- Rejected: None of the definitions match the benchmark's definitions at all.

5.3 Results

The ChengyuSTS dataset is prohibitively time-consuming for manual annotation. Due to this, only 1005 entries out of 25753 were sampled.

5.3.1 Detection Metrics. As illustrated in Table 1, our translator could successfully identify most sentences that contain an idiom, as well as detect its location in the sentence. It never hallucinated an idiom when it did not exist, and could very safely ignore sentences that did not have idioms. However, there is a sizable number of sentences that contained idioms that were not detected by our translator.

Condition	**Count**
Idioms, Successfully Detected (TP)	560
No idioms, Successfully Detected (TN)	250
Idioms, Unsuccessfully Detected (FN)	195
No idioms, Falsely Detected (FP)	0

Table 1. Number of idioms given if they were present in a sentence and if they were detected.

Precision	1.00
Recall	0.742
F1-Score	0.852
Accuracy	0.806

Table 2. Detection metrics calculated from Table 1

By labeling each condition of idioms and their detection as True Positives, True Negatives, False Negatives, and False Positives respectively, we are able to calculate Precision, Recall, F1-Score, and Accuracy. The high precision shows that our detection is highly reliable when it is able to detect an idiom with a sentence, but the lower recall suggests that its detection ability is slightly lacking in comparison.

5.3.2 Definition Metrics. The next two tables display results from comparing the definitions from the `/translate` endpoint to the benchmark definitions from GPT-5.4 via a human expert.

Label	**Count**
Approved	499
Rejected	15
Unsatisfactory	46

Table 3. Labels of idiom definitions from the pool of sentences that were successfully identified to contain an idiom.

Strict Accuracy	0.891
Lenient Accuracy	0.973
Weighted Accuracy	0.932

Table 4. Accuracy of the translator's idiom definitions of varying strictness.

The results from Table 3 indicate that most idioms that our translator expressed were synonymous with the benchmark's definitions. To put it into further perspective, we can observe the varying degrees of accuracies as shown in Table 4. Strict Accuracy only takes into account of only Approved definitions over the 560 definitions that were calculated; Lenient Accuracy also takes the Unsatisfactory labels into consideration as well, while Weighted Accuracy takes a more balanced approach - assigning a weights system of 1.0 points to Approved labels, and 0.5 points to Unsatisfactory labels. Even if we were to evaluate our definition capability in the lens of Strict Accuracy, it could still be considered as acceptable, though could still be improved.

6 Conclusion

In this project, we developed a Chinese–English idiom-aware translation system designed to improve contextual understanding of idiomatic expressions. Unlike traditional machine translation systems that often produce literal translations, our approach combines translation APIs with embedding-based idiom retrieval and contextual explanation generation to better preserve semantic and cultural meanings.

The system integrates a React frontend, a Node.js Express/FastAPI backend, and a FAISS-based semantic retrieval pipeline. By aligning the PETCI and IdiomTranslate30 datasets, the system is able to detect idiomatic expressions, provide contextual English interpretations, and display bilingual example sentence pairs and explanations to users.

Our implementation demonstrates that combining semantic retrieval with existing translation systems can improve the interpretability and usability of machine translation for idiomatic expressions. The project also highlights the importance of contextual and cultural information in cross-lingual communication and language learning applications.

In future work, the system could be extended with larger multilingual idiom datasets, improved embedding models, and LLM-based explanation generation to support more accurate and flexible idiom

understanding. Other possible implementations include computer vision integration or packaging the translator into a browser application or extension can further reduce the friction of the user interaction and translating text even further, or extending the system to reverse the functionality so as to translate English idioms in concise and understandable Chinese.

7 Team Contributions

Abuzar Hussain Mohammad was responsible for building FAISS index pipeline, and idiom detection and extraction module.
Yixin Zhou was responsible for building idiom dataset and sentence translation module.
Larry Liao was responsible for developing the idiom translation frontend and integrating the various modules into it.
Andrew Chen was responsible for developing the evaluation methodology pipeline as well as delivering it.
Final report and presentation preparation are evenly shared across team members.

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